

# TeamSync — Database Schema & ER Specifications

Relational Data Models, Constraints & Field Specifications | Author: Rajkumar PR

## 1. Relational Entity Architecture Overview

The TeamSync database architecture consists of 7 normalized relational tables enforcing referential integrity via foreign keys, cascading deletions, unique index constraints, and timestamp auto-recording.

| Model Name        | Django Model Class  | Primary Function                                                        |
|-------------------|---------------------|-------------------------------------------------------------------------|
| User              | User (AbstractUser) | Stores system credentials, roles (ADMIN/MEMBER), and department         |
| Project           | Project             | Tracks project scope, dates, status, creator, and assigned members      |
| ProjectInvitation | ProjectInvitation   | Manages join requests (PENDING/ACCEPTED/REJECTED) sent to members       |
| Task              | Task                | Stores task title, priority, status, deadline, and assigned team member |
| DeadlineHistory   | DeadlineHistory     | Audit log recording previous vs new deadline, justification, and editor |
| Comment           | Comment             | Discussion comments and task status progress updates                    |

## 2. Detailed Table Field Specifications

Table 2.1: User (users\_user)

| Field      | Data Type    | Constraints        | Description           |
|------------|--------------|--------------------|-----------------------|
| id         | INTEGER      | PK, AUTO_INCREMENT | Unique user ID        |
| username   | VARCHAR(150) | UNIQUE, NOT NULL   | Login username        |
| email      | VARCHAR(254) | UNIQUE, NOT NULL   | User email address    |
| role       | VARCHAR(20)  | DEFAULT 'MEMBER'   | Role: ADMIN or MEMBER |
| department | VARCHAR(100) | NULLABLE           | User department       |

Table 2.2: Project (projects\_project)

| Field         | Data Type    | Constraints           | Description                       |
|---------------|--------------|-----------------------|-----------------------------------|
| id            | INTEGER      | PK, AUTO_INCREMENT    | Unique project ID                 |
| name          | VARCHAR(255) | NOT NULL              | Project title                     |
| status        | VARCHAR(20)  | DEFAULT 'ACTIVE'      | PLANNING/ACTIVE/ON_HOLD/COMPLETED |
| created_by_id | INTEGER      | FK(User.id), SET_NULL | Admin creator ID                  |

Table 2.3: Task (tasks\_task)

| Field          | Data Type    | Constraints             | Description                          |
|----------------|--------------|-------------------------|--------------------------------------|
| id             | INTEGER      | PK, AUTO_INCREMENT      | Unique task ID                       |
| title          | VARCHAR(255) | NOT NULL                | Task title                           |
| status         | VARCHAR(20)  | DEFAULT 'TODO'          | TODO/IN_PROGRESS/IN_REVIEW/COMPLETED |
| priority       | VARCHAR(20)  | DEFAULT 'MEDIUM'        | LOW/MEDIUM/HIGH/URGENT               |
| deadline       | DATETIME     | NOT NULL                | Task target deadline                 |
| project_id     | INTEGER      | FK(Project.id), CASCADE | Parent project ID                    |
| assigned_to_id | INTEGER      | FK(User.id), SET_NULL   | Assigned team member ID              |

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